

Test Driven Development

Best Coding Practices

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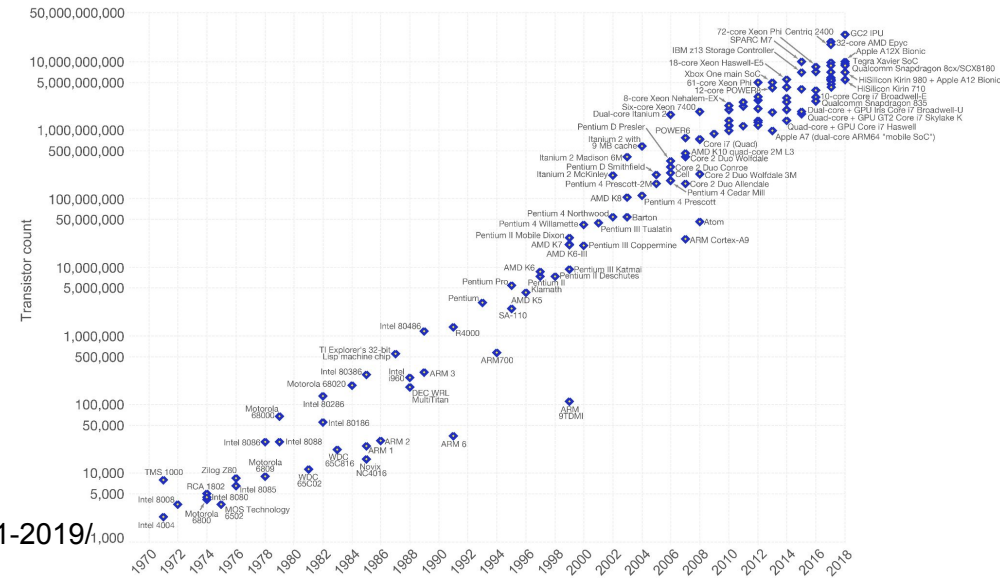
DSR 2025

**Q: How many computers are there
on a spaceship?**

The Random Neutrino Problem

- Voting Machine changed outcome because of random collision with a cosmic particle
- Mario jumps in a way that the game did not plan for
- A plane almost crashed!

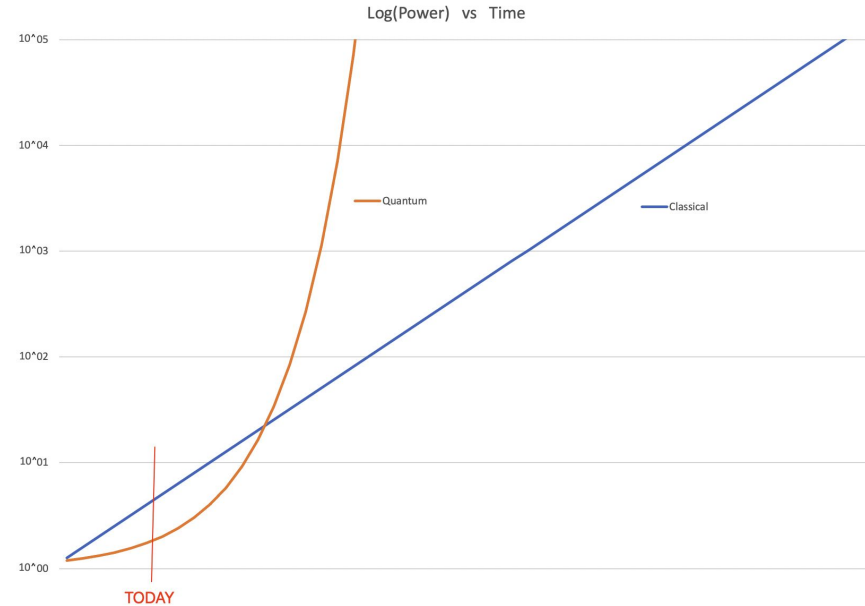
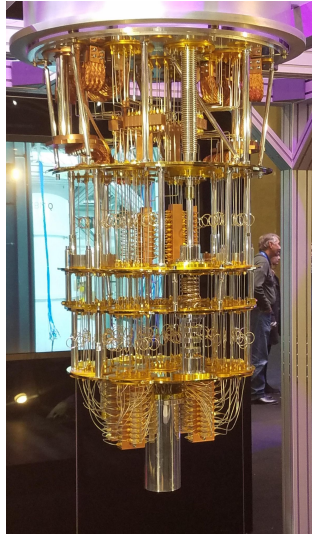
● Moores Law and Quantum Computers



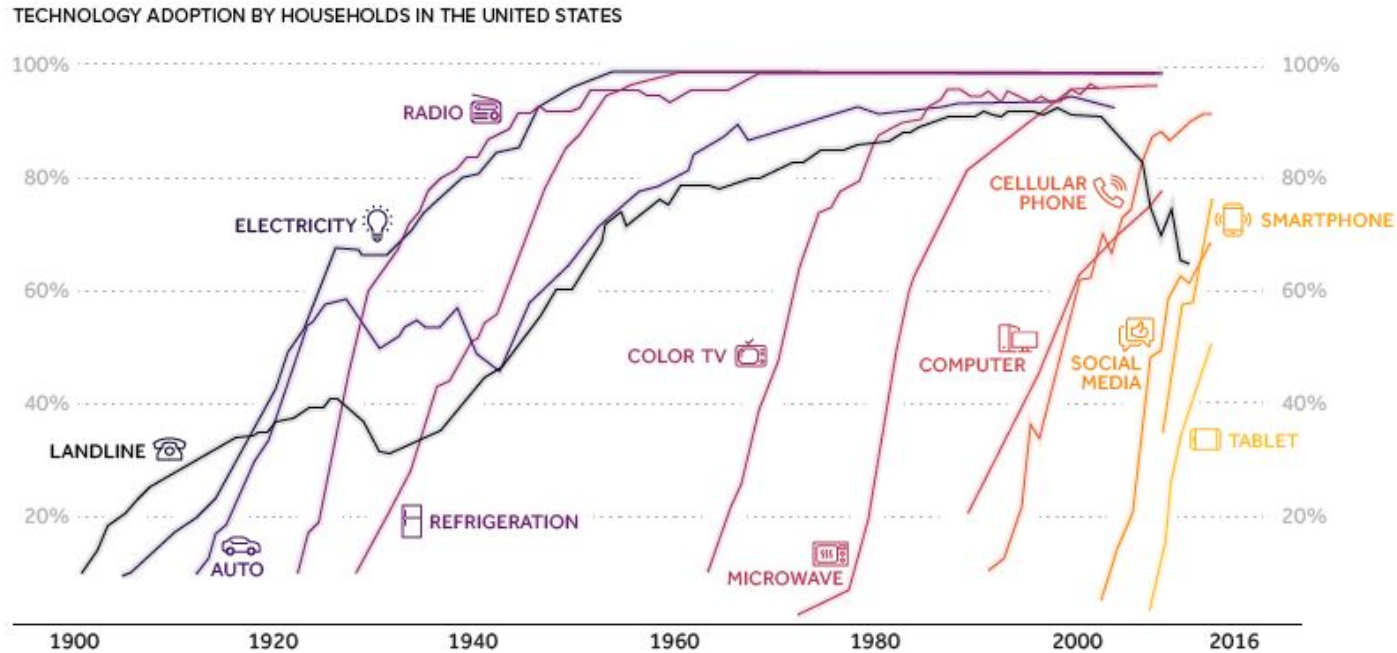
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Endless Loops and Finite Resources

- Quantum Computers (IBM Q)



Technology adaptation and range

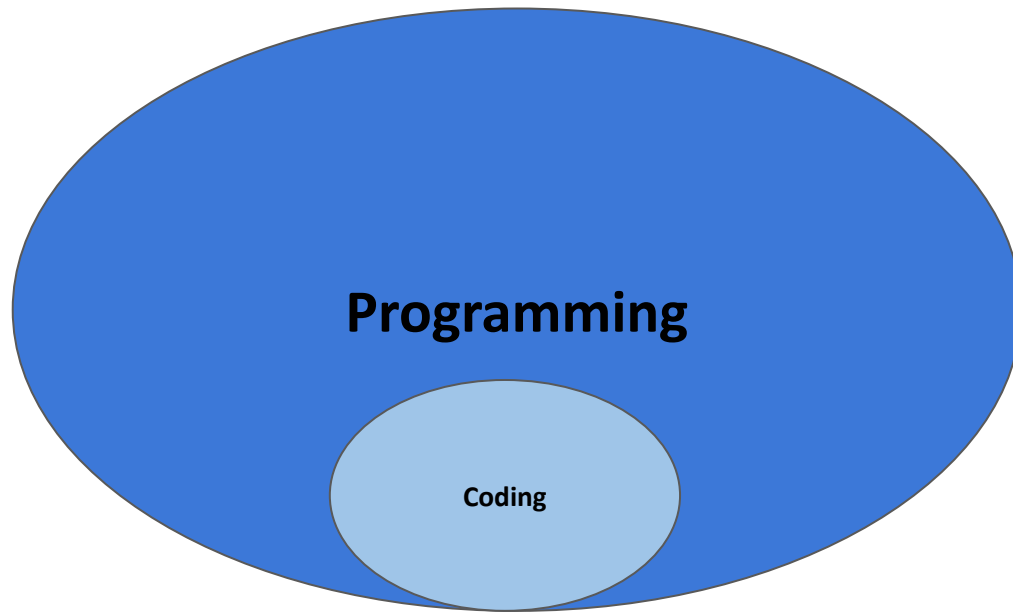


Coding and Programming

Q: What is the difference?

The most important part is...

HAVING A PLAN!



A: 'Physical activity of writing code'

Programming as a Concept

With many equally important parts!

- **Planning!**
- Problem Solving
- Design
- Collaboration
- Innovation
- Long Term Thinking
- Organization
- ... Actually writing some code! :D

Tests: And why do we NEED them.

1. Developer writes code
2. Git add, git commit, git push, merged to main
3. Maintainer/Reviewer is on vacation
4. Code goes to production
5. Code doesn't work
6. App breaks
7. No profit (actually the opposite of profit... loss!)
8. Maintainer/Reviewer disturbed on a faraway tropical island in the middle of the pacific while freediving with whales, the freshly made pina colada drops from his hand and shatters on the sun-bathed rocks...

Tests: And why do we NEED them?

1. Developer writes some code
2. Git add, git commit, git push—test goes red!
3. Merge reverted!
4. App still works!
5. Profit!
6. Maintainer/Reviewer happily enjoys his vacation

Tests - Practical Examples from the industry

- Quality Assurance and Bug Fixing
- Testers and Developers are two separate teams writing completely different code!
- Developers don't like testers, as their job is to break the code of the devs
- Testers think that developers are lazy bums



Tests: Practical Examples from the industry

A software tester goes into a bar.

Orders a beer.

Orders 0 beers.

Orders 999999999 beers.

Orders a bear.

Orders -1 beers.

Orders hdtseatfibkd.

The first real customer walks into a bar and asks where the bathroom is.

The bar bursts into flames, killing everyone inside.

Tests: Practical Examples from the industry

All companies are testing.

But some can afford not to test in production!

It is a little bit like early-access games

- Pay us before official release
- So we don't have to pay testers
- ... profit?



Tests: Practical Examples from the industry

Tested mindset:

- Finding holes that were omitted in development
- When a test fails, quickly points a developer to the cause
- Making sure the test coverage is as complete as possible
- Foolproof the app
- Create usage scenarios that reveal weaknesses
- Keep the application working in development and after release
- Make sure the app will pass certifications

Tests: Practical Examples from the industry

Tests to Pass

- Plan the functionality and define requirements
- Project indicators are clearly defined before the project starts
- Write the test structure with the final functionality in mind
- Developer add functionality until all tests are passed

Tests to Fail

- Code tested already exists and/or is on a completely different platform:
 - E.g Wireless communication with other devices
- Certain functionalities need to pass standards (for certification)
- Catching bugs

TDD – Test Driven Developement

- We are writing tests to pass!
- Create the test structure beforehand
- Write code until You pass!

This has some setbacks:

- Rigid test structure often obstructs functionality
- Lack of flexibility
- Stacking tests can result in a mess

Is there an alternative?

- Do not write any tests ;p
 - Better have SOME rigid tests than none at all
- **BDD :o**

BDD - Behaviour Driven Development

- Focus on behavior or interactions of the system
- Human readability
 - Scenarios!
- Given-When-Then
 - **Given:** Preconditions or initial context.
 - **When:** Actions or events.
 - **Then:** Expected outcomes or results.

Technologies:

- Pytest-bdd
- Behave
- Lettuce
- gherkin

BDD – Behaviour Driven Developement

- Gherkin

Feature: Addition

Scenario: Add two numbers

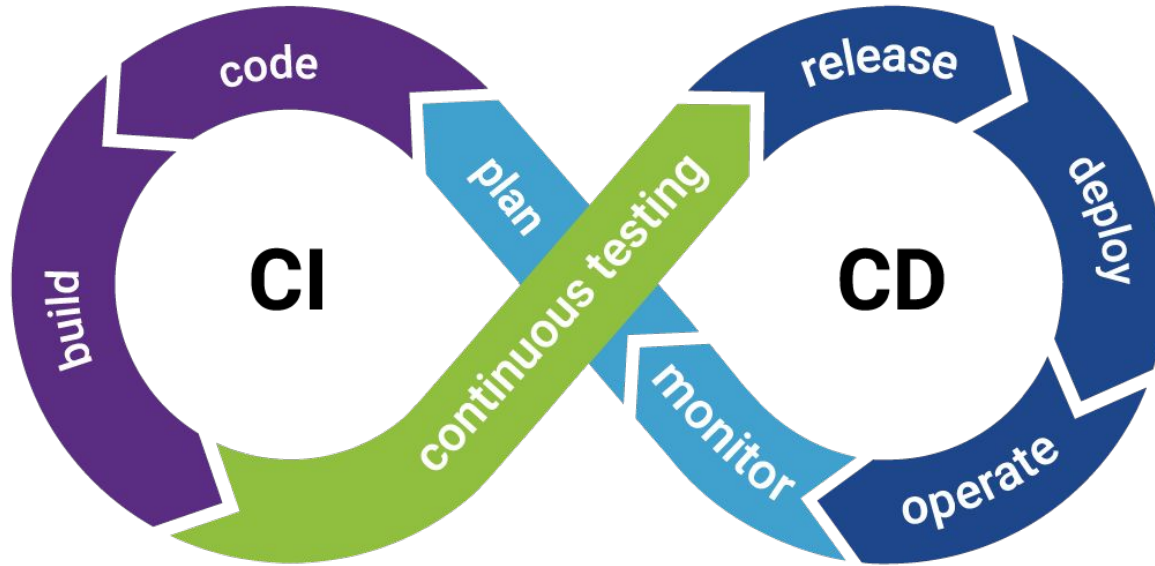
Given I have two numbers 2 and 3

When I add them together

Then I should get the result as 5

CI/CD - DevOps

- Continuous Integration / Continuous Delivery



CI/CD - DevOps

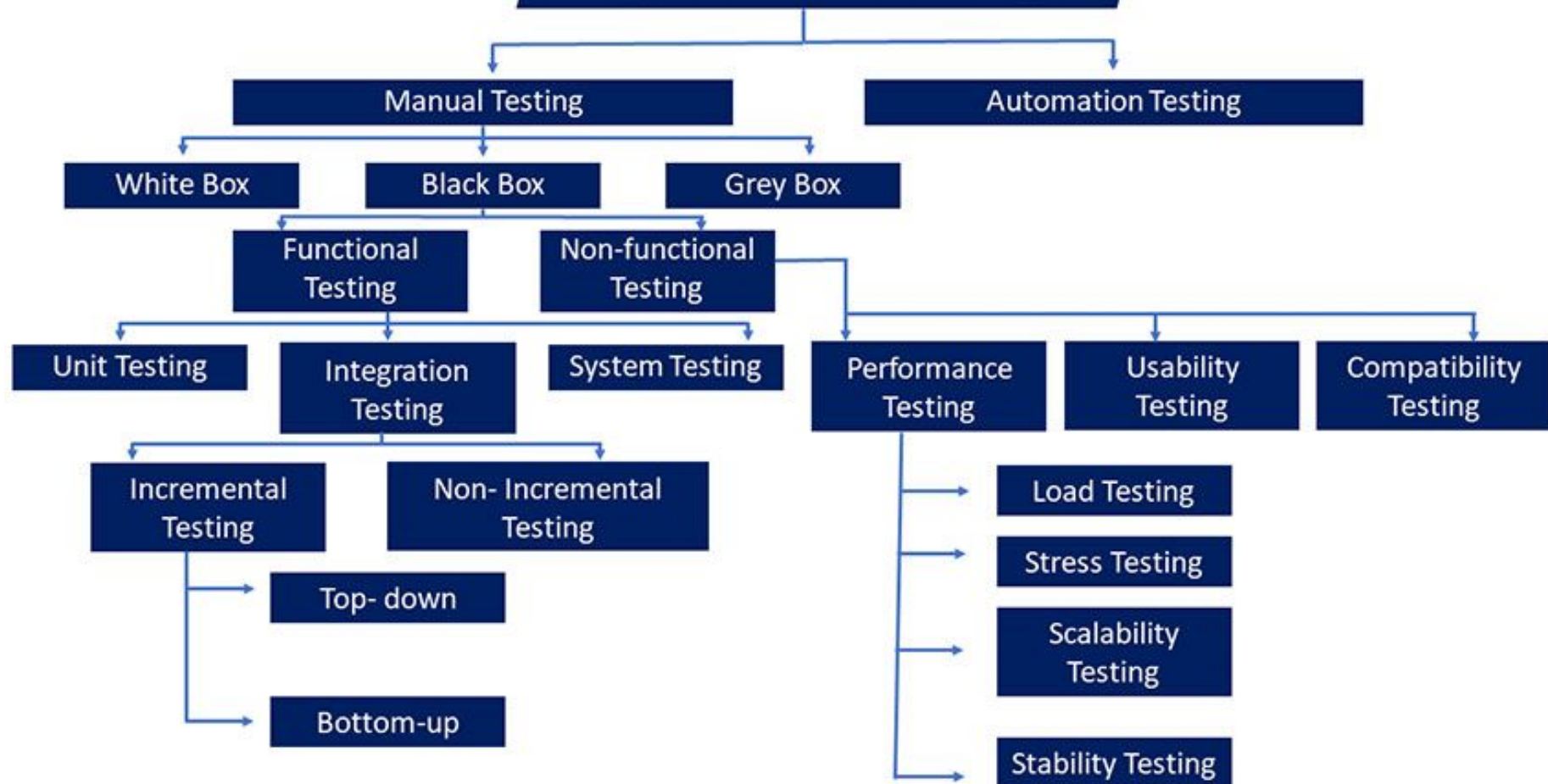
- Funny integrated in GitHub and GitLab
- Automatic scripting applied to branches (usually master or release)
- Script runs stages e.g.:
 - Setup
 - Build
 - Test
 - Deploy
- The scripts execution is done before merging with production

If anything goes bad, merge is reverted!

ISTQB

- International Software Testing Qualification Board
 - Definitions of test types and levels
 - Best practices and practical examples of test coverage
 - Agile test development
 - Moderately priced Certification **(this is not an Ad)**
 - Good way into first python development job is to start in QA (testing)
 - Knowledge of ISTQB is very important for that!

Types of Software Testing



Unittest or pytest?

Well they are different things!

- Unit test is great for unit testing (as the name suggests), but it is not so popular with anything more.
- Unittest is included with any modern python distribution
- Amazing for testing singular modules, objects or functions if they work correctly

Unittest or pytest?

- Pytest is an external library, but become dominantly popular
- It is convenient and flexible
- Can perform unit tests and integration tests at any complexity
- Highly customizable with fixtures
- Compatible with many proprietary repositories (like Allure)

Tests to Fail Practical - Meter Mystery!

- You are given a water meter that communicates its status wirelessly
- Your job is to check if the meter will pass certification
- You are given requirements for the certification
- Write any tests You deem necessary to check if meter could be certified
- Report any discrepancies from the requirements

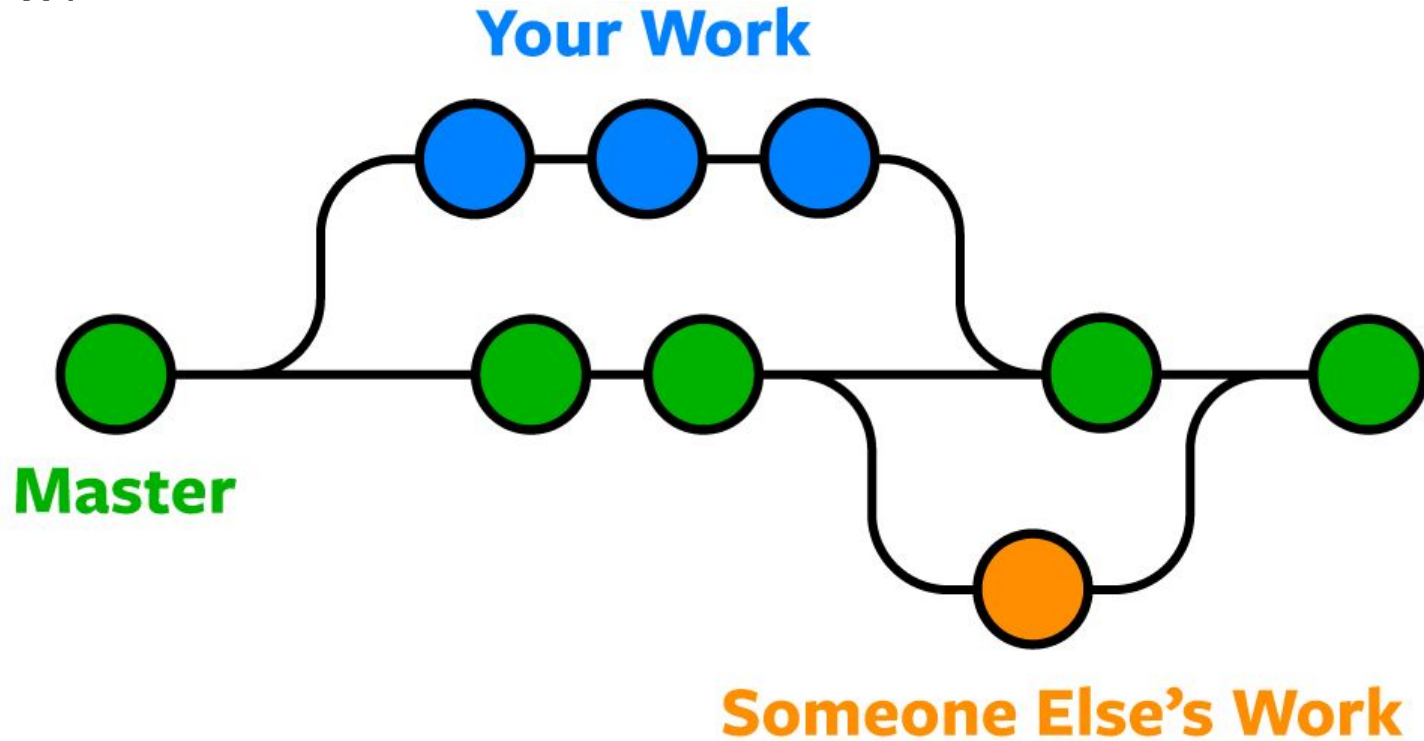
**Write me a private message on Slack if You think
You got it all!**

Practical aspects of Project Management

- Version Control
 - Git
 - Git GUIs
 - GitHub
 - GitLab
- Agile
 - Jira/Kanban/Scrum
 - Notion
- Project Structure from Scratch
 - [Example project structure](#)
 - Cookiecutter!

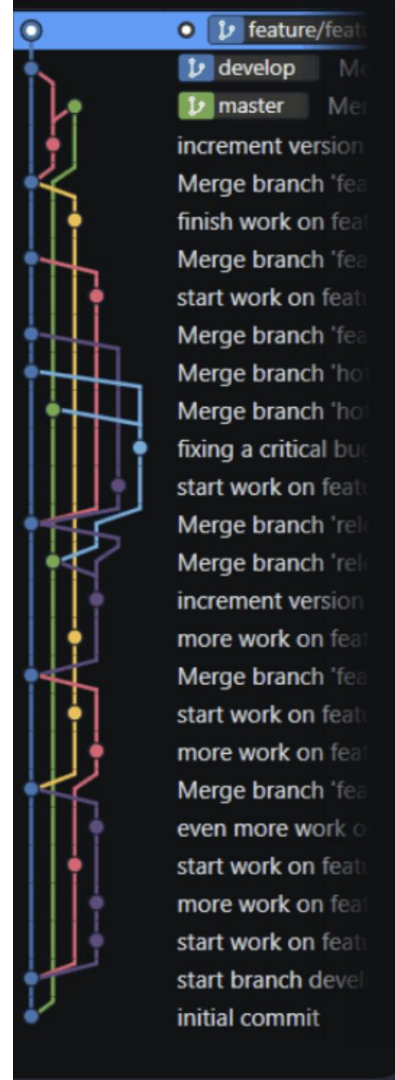
Version control

-
-
-



Version control

- Git visual APIs
 - Fork (Windows, Mac)
 - GitKraken (Ubuntu)
 - Great PyCharm Integration!



Agile

- **Jira**
 - Product by Atlassian
 - Uses both Scrum and Kanban methodologies
 - Interactive task management tool
- **Scrum**
 - Workflow is time-boxed (Sprints)
 - Specific roles
 - Designed to regulate workflow and ensure deployment efficiency
- **Kanban**
 - Not regulated work-time
 - No specific roles
 - Flexibility and adapting to capacity to keep the workflow moving

All are visual tools to help productivity and task optimization

Day Two, End of Daily Scrum





Teams in Space

Software project



Backlog



Board



Reports



Releases



Components



Issues



Repository



Add item



Settings



Board

Release



Quick filters ▾

Assignee ▾

Sprint ▾

TO DO 5

IN PROGRESS 5

IN REVIEW 2

DONE 8

▽ Super important 11 issues

When requesting user details the service should return prior trip info

SEESPACEEZ PLUS



TIS-37

Engage Jupiter Express for outer solar system travel

SPACE TRAVEL PARTNERS



5

TIS-25



Create 90 day plans for all departments in the Mars Office

LOCAL MARS OFFICE



9

TIS-12

Requesting available flights is now taking > 5 seconds

SEESPACEEZ PLUS



3

TIS-8



Engage Saturn Shuttle Lines for group tours

SPACE TRAVEL PARTNERS



4

TIS-15



Establish a catering vendor to provide meal service

LOCAL MARS OFFICE



4

TIS-15



Register with the Mars Ministry of Revenue

LOCAL MARS OFFICE



3

TIS-11

Draft network plan for Mars Office

LOCAL MARS OFFICE



3

TIS-15



Add pointer to main css file to instruct users to create child themes

LARGE TEAM SUPPORT



TIS-56

Homepage footer uses an inline style - should use a class

LARGE TEAM SUPPORT



TIS-68



Engage JetShuttle SpaceWays for travel

SPACE TRAVEL PARTNERS

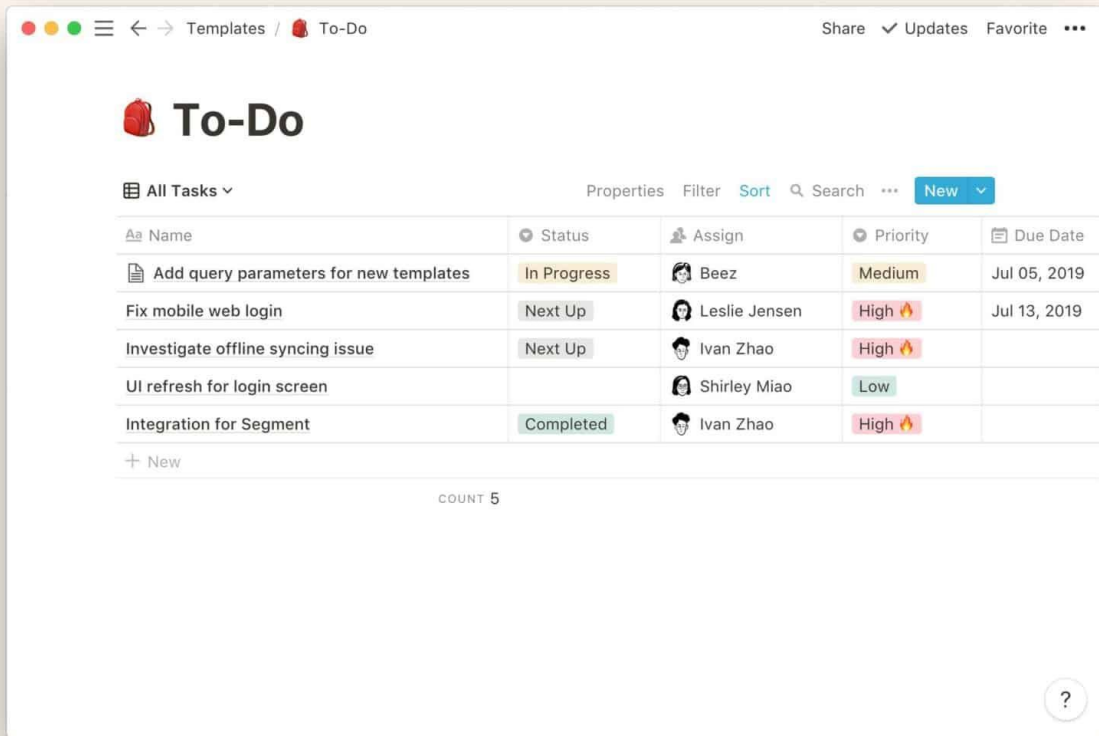


5

TIS-23



Notion - All in one



The screenshot shows a Notion workspace titled "To-Do" with a red backpack icon. The interface includes a top navigation bar with "Templates / To-Do", "Share", "Updates", and "Favorite" options. Below the title, there are tabs for "All Tasks" (selected), "Properties", "Filter", "Sort", "Search", and a "New" button. The main content is a table with five columns: Name, Status, Assign, Priority, and Due Date. The table contains six tasks, with the first one expanded to show a description. The tasks are: "Add query parameters for new templates" (In Progress, Beez, Medium, Jul 05, 2019), "Fix mobile web login" (Next Up, Leslie Jensen, High, Jul 13, 2019), "Investigate offline syncing issue" (Next Up, Ivan Zhao, High), "UI refresh for login screen" (Shirley Miao, Low), and "Integration for Segment" (Completed, Ivan Zhao, High). A "+ New" button is at the bottom left, and a "COUNT 5" label is at the bottom center. A help icon (?) is in the bottom right corner.

Name	Status	Assign	Priority	Due Date
Add query parameters for new templates	In Progress	Beez	Medium	Jul 05, 2019
Fix mobile web login	Next Up	Leslie Jensen	High	Jul 13, 2019
Investigate offline syncing issue	Next Up	Ivan Zhao	High	
UI refresh for login screen		Shirley Miao	Low	
Integration for Segment	Completed	Ivan Zhao	High	

Let's do some co.. Programming!

GitLab Project Showcase

- Create a Project on GitHub
- Project Management
- GitHub Actions - CICD and Project Automation
- Deploying Documentation with Sphinx

